

DSA (Abdul Bari) - Day 6

45-day plan | YouTube playlist course

Arrays - Basics

Contiguous memory, $O(1)$ indexing, fixed vs dynamic.

Key concepts

- Access by index is $O(1)$; insert/delete in middle is $O(n)$.
- Static vs dynamic arrays (resizing).
- Row-major memory layout.

Key idea / example

```
arr[i] address = base + i * size -> constant-time access.
```

Today's tasks (1.5h)

- ☐ List array operations & costs
- ☐ Implement insert/delete
- ☐ Note dynamic resizing

Resources

Playlist: youtube.com/playlist?list=PLsr8vTgyLdy_YndxNcI4WkH5Vorj5qvrv

(Clickable playlist link is in the app: Goals > this day > Study block.)